

2. (a) (i)  $V_0$  = total volume of air required at  $P_0$ , including the residual air in the ball

$$P_1 V_1 = P_0 V_0$$

$$(156 \text{ kPa}) (6000 \text{ cm}^3) = (100 \text{ kPa}) (V_0)$$

$$V_0 = 9360 \text{ cm}^3$$

$$\begin{aligned} \therefore \text{volume of air} &= V_0 - \text{volume of the basketball} \\ &= 9360 \text{ cm}^3 - 6000 \text{ cm}^3 \\ &= 3360 \text{ cm}^3 \end{aligned}$$

1M

1A

1M

Alternatively:

$$\therefore n = \frac{PV}{RT}$$

$$\frac{156V}{RT} - \frac{100V}{RT} = n'$$

$$\therefore V' = \frac{56V}{RT} \cdot \frac{RT}{100} = 3360 \text{ cm}^3$$

1M

1M + 1A

3

- (ii) Number of strokes required =  $3360 \text{ cm}^3 \div 120 \text{ cm}^3$   
= 28

1A

1

- (b) Using kinetic theory, the pressure  $p$  is given by

$$pV = \frac{N m \overline{c^2}}{3} = \frac{2N}{3} \cdot \frac{m \overline{c^2}}{2}$$

As the volume and the temperature ( $\propto$  kinetic energy of the air molecules) remain unchanged,  
the increase in pressure is due to the increase of number of air molecules hitting the wall of the container per unit time.

1A

1A

Or Frequency of collision with the wall of container increases.

1A

2

| Solution                                                                                                                                                                 |                                                                                                                                                                                                                                            | Marks                                      | Remarks |                        |          |                   |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------|------------------------|----------|-------------------|-------|
| 5. (a)                                                                                                                                                                   | (i) $m(5.0 \text{ cm}) = 50 \text{ g}$ (10.0 cm)<br>$m = 100 \text{ g}$ or $0.1 \text{ kg}$                                                                                                                                                | 1M<br>1A                                   |         |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                            | 2                                          |         |                        |          |                   |       |
|                                                                                                                                                                          | (ii) Counter-weight position : $10.0 \text{ cm} \pm 0.1 \text{ cm}$<br>Percentage error = $100\% \times \left(\frac{0.1}{10.0}\right) = 1\%$<br>$\therefore m = 101 \text{ g}$ to $99 \text{ g}$<br>i.e. maximum error = $\pm 1 \text{ g}$ | 1M<br>1A                                   |         |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                            | 2                                          |         |                        |          |                   |       |
|                                                                                                                                                                          | (b) Spring balance reading = $mg = (0.1 \text{ kg})(9.81 \text{ N kg}^{-1})$<br>$= 0.981 \text{ N}$ ( $1.0 \text{ N}$ for $g = 10 \text{ N kg}^{-1}$ )<br>$\approx 1.0 \text{ N}$                                                          | 1M/1A                                      |         |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                            | 1                                          |         |                        |          |                   |       |
|                                                                                                                                                                          | (c) (i)                                                                                                                                                                                                                                    |                                            |         |                        |          |                   |       |
|                                                                                                                                                                          | <table border="1"><tr><td>counter-weight position<br/>on beam balance</td><td>spring balance reading</td></tr><tr><td>the same</td><td>reading increases</td></tr></table>                                                                 | counter-weight position<br>on beam balance |         | spring balance reading | the same | reading increases | 1A+1A |
|                                                                                                                                                                          | counter-weight position<br>on beam balance                                                                                                                                                                                                 | spring balance reading                     |         |                        |          |                   |       |
|                                                                                                                                                                          | the same                                                                                                                                                                                                                                   | reading increases                          |         |                        |          |                   |       |
|                                                                                                                                                                          | 2                                                                                                                                                                                                                                          |                                            |         |                        |          |                   |       |
| (ii) The beam balance would fail to work / to measure the mass of the load,<br>as the apparent weight is zero (or weightless), the counter-weight can take any position. | 1A<br>1A                                                                                                                                                                                                                                   |                                            |         |                        |          |                   |       |
|                                                                                                                                                                          | 2                                                                                                                                                                                                                                          |                                            |         |                        |          |                   |       |